

You have a chip antenna & want to use it at 300 MHz. At this frequency you find it has an impedance of $25 - 150j \Omega$

Design an L match for $Z_0 = 50 \Omega$.

① Add the reflected matching circle.

② Normalize & plot. $\Rightarrow Z = 0.5 - 3j$
($\div 50$)

③ Consider network. Z is impedance so series & we need to add j to get to the circle. So we need a series inductor to add $2.5j$ $Z \rightarrow A$ or $3.5j$ $Z \rightarrow B$

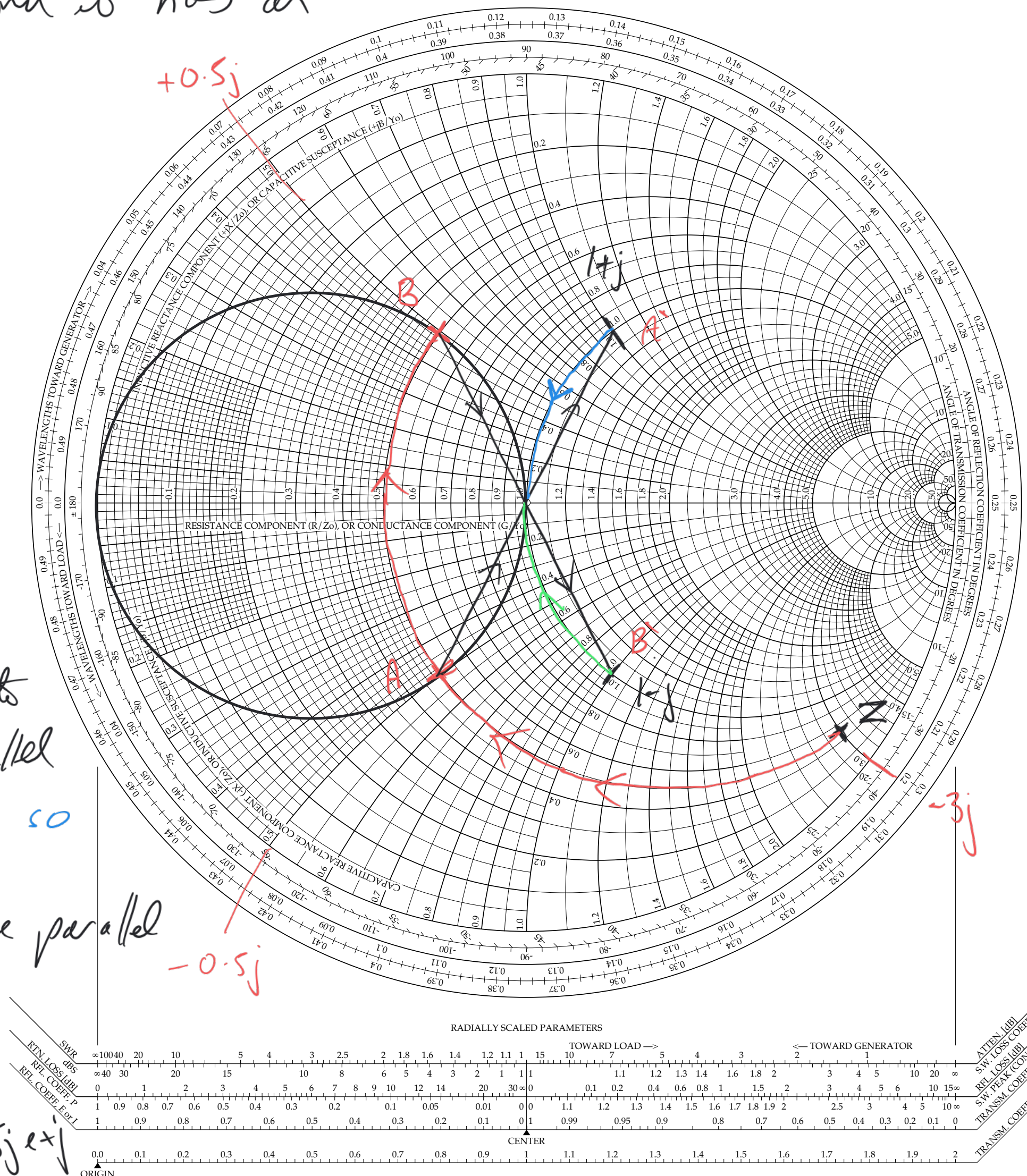
④ If A then we transpose through circle to same distance the other side so in parallel & admittance. We need to remove $1j$ so use a parallel inductor

If B we transpose again to admittance & parallel
Now we need to add j so use a parallel capacitor.

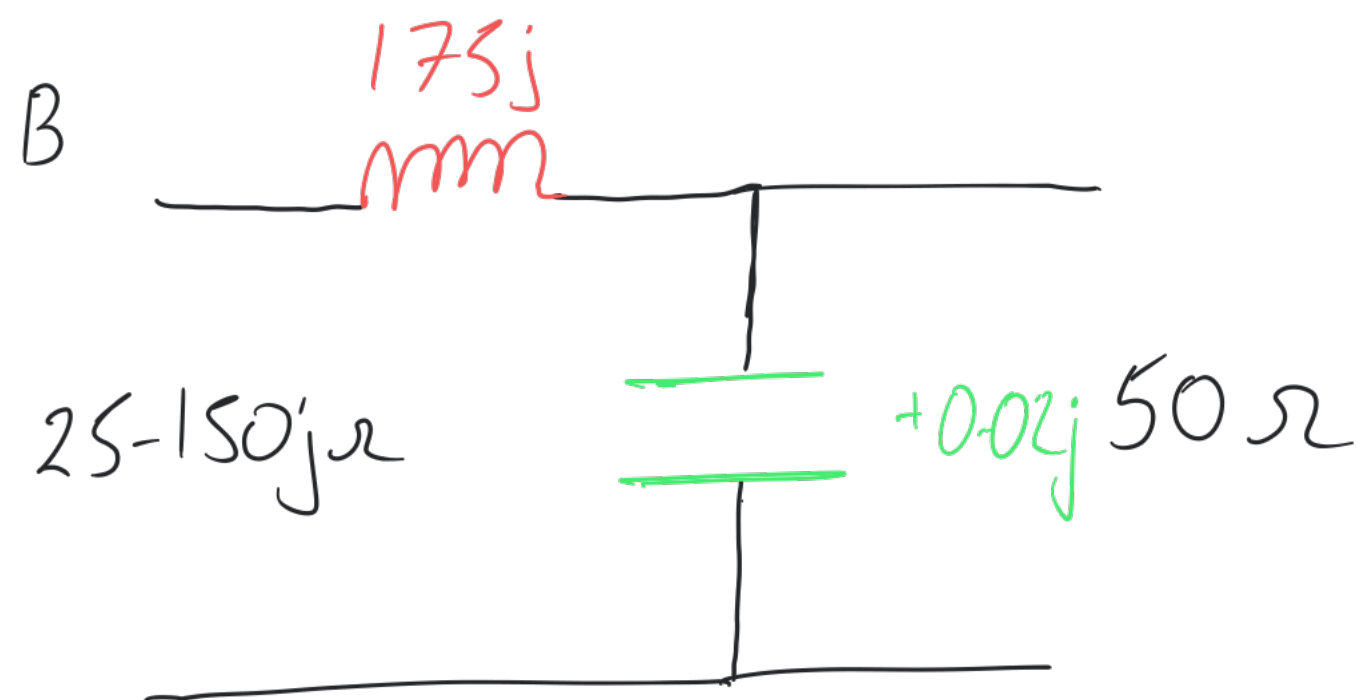
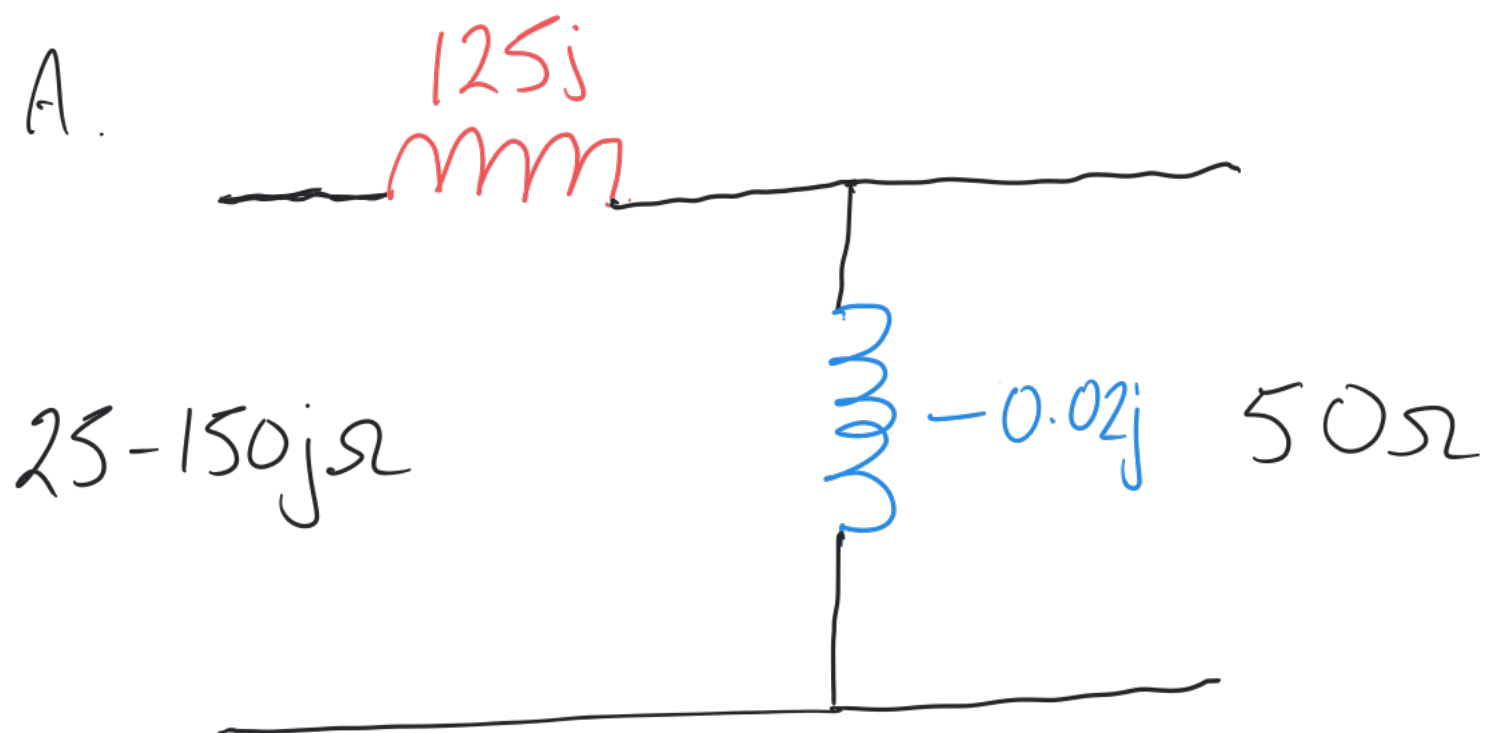
Value from chart $A = +2.5j$ & $-j$ $B = +3.5j$ & $+j$

The Complete Smith Chart

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Putting our two networks together: $A = 2.5j e^{-j}$ $B = 3.5j e^j$
 De-Norm = $125j e^{-0.02j}$ $175j e^{0.02j}$



Values at 300MHz $\omega = 2\pi f$.

A

$$X_L = 125j = j\omega L$$

$$\frac{125j}{j\omega} = L = \underline{\underline{66 \text{ nH}}}$$

$$Y_L = -0.02j = \frac{1}{j\omega L}$$

$$L = \frac{1}{-j\omega 0.02j} = \frac{1}{\omega 0.02} = \underline{\underline{26 \text{ nH}}}$$

B

$$X_L = 175j = j\omega L \quad L = \frac{175}{\omega} = \underline{\underline{92 \text{ nH}}}$$

$$Y_C = 0.02j = j\omega C \quad C = \frac{0.02j}{j\omega} = \underline{\underline{10 \text{ pF}}}$$

Remember

Series

Ω		$+j$ (up chart)	$X_L = j\omega L$	Onbo chart (Normalise) $\div 50$	Off chart (de. normalize) $\times 50$
ohms Ω		$-j$ (down chart)	$X_C = \frac{1}{j\omega C}$	$\div 50$	$\times 50$

Parallel

Ω		$-j$ (down chart)	$Y_L = \frac{1}{j\omega L}$	$\times 50$	$\div 50$
siemens. Ω		$+j$ (up chart)	$Y_C = j\omega C$	$\times 50$	$\div 50$

The input to a MOSFET amplifier needs match
at $f = 300 \text{ MHz}$. Its input admittance is:

$$Y_L = 0.004 + 0.02j \text{ S}$$

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① Add the reflected matching circle.

② Normalize & plot.

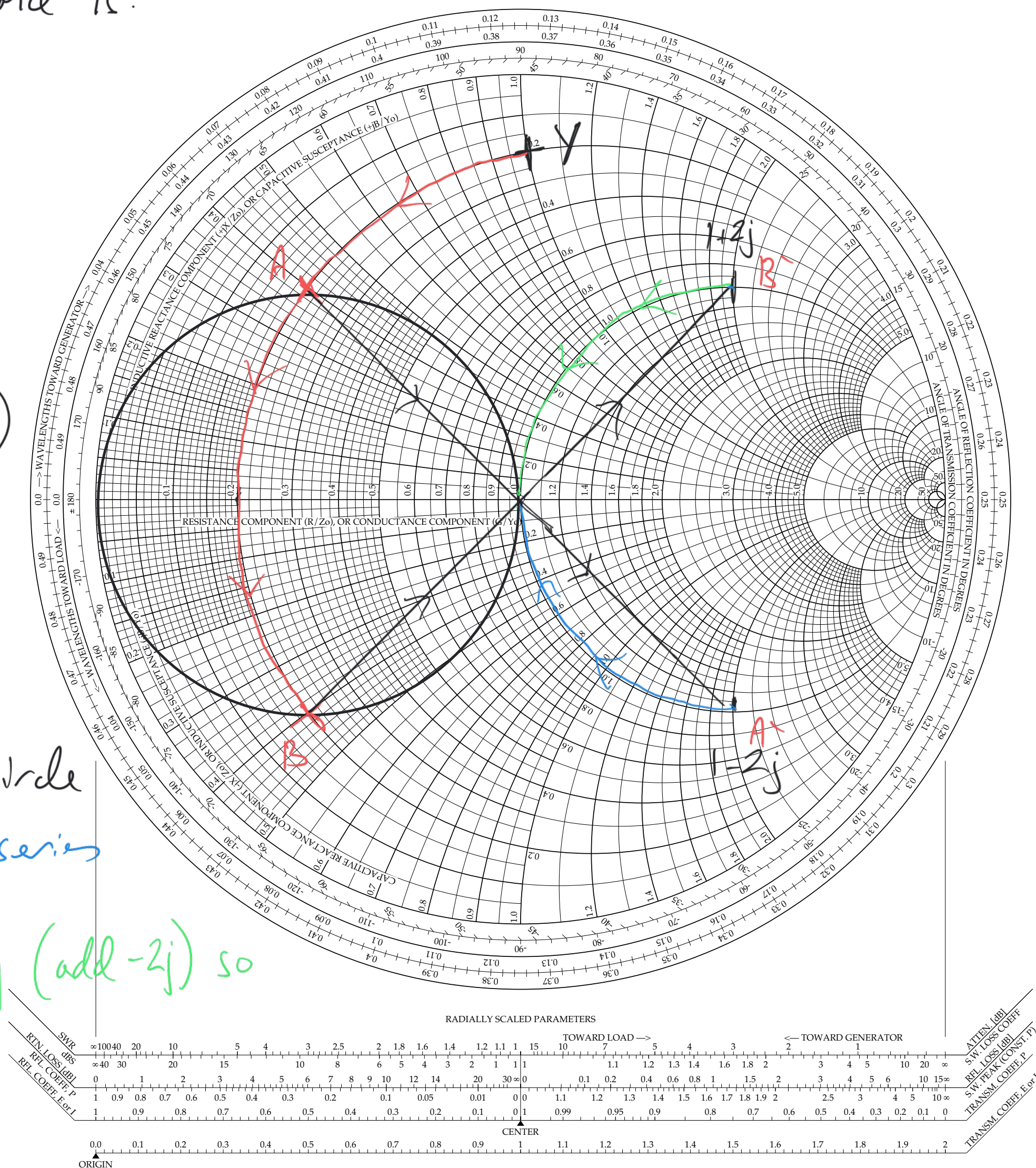
$$\text{So admittance} = \times 50.$$

$$\Rightarrow Y = 0.2 + j$$

③ We need to subtract j (add $-j$)
to get down to circle. We are
admittance/parallel so need an
inductor. Either $A - 0.6j$
or $B - 1.4j$

④ Transpose to series onto matching circle
For A' we need to add $2j$ so series
inductor.

For B' we need to subtract $2j$ (add $-2j$) so
series capacitor.



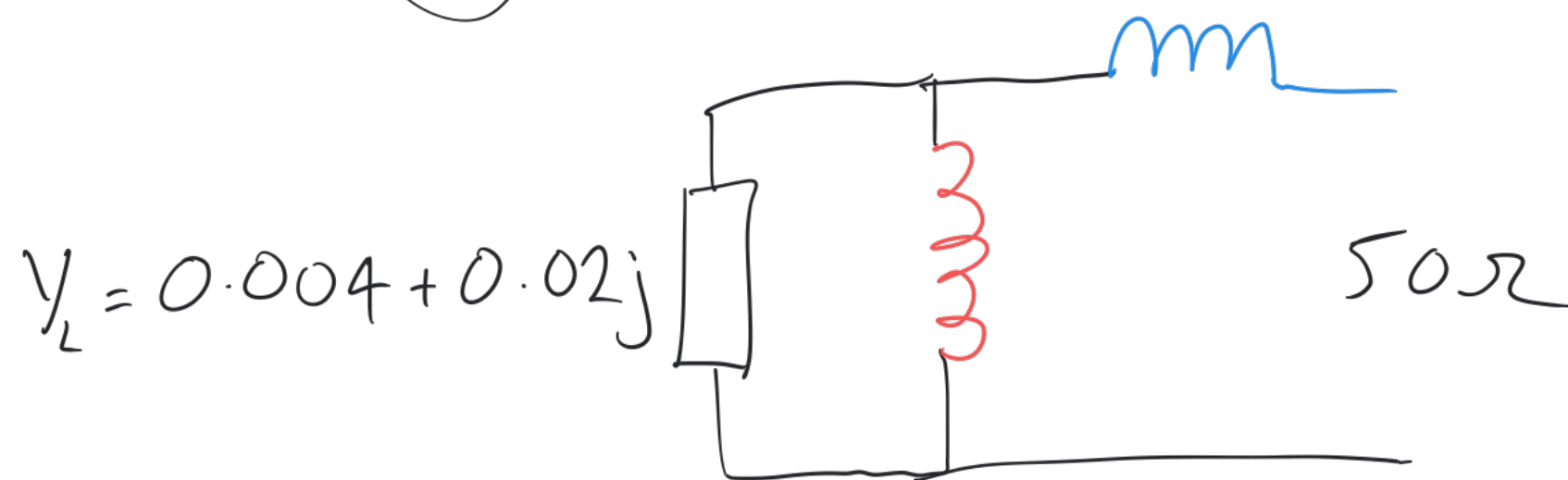
Our two choices

(A)

$$Y_L = -0.6j \quad X_C = +2j$$

De-normalize $Y_C = -0.012j \quad X_L = 100j$

Calculate actual values

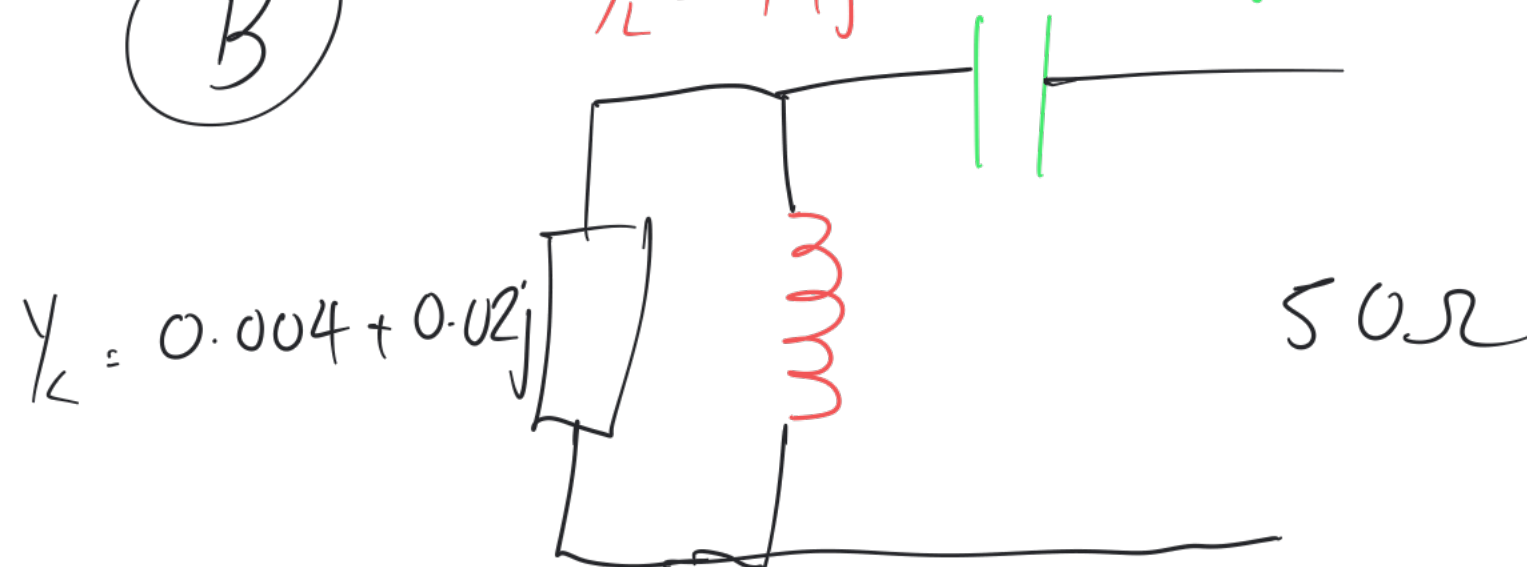


(B)

$$Y_L = -1.4j \quad X_C = 2j$$

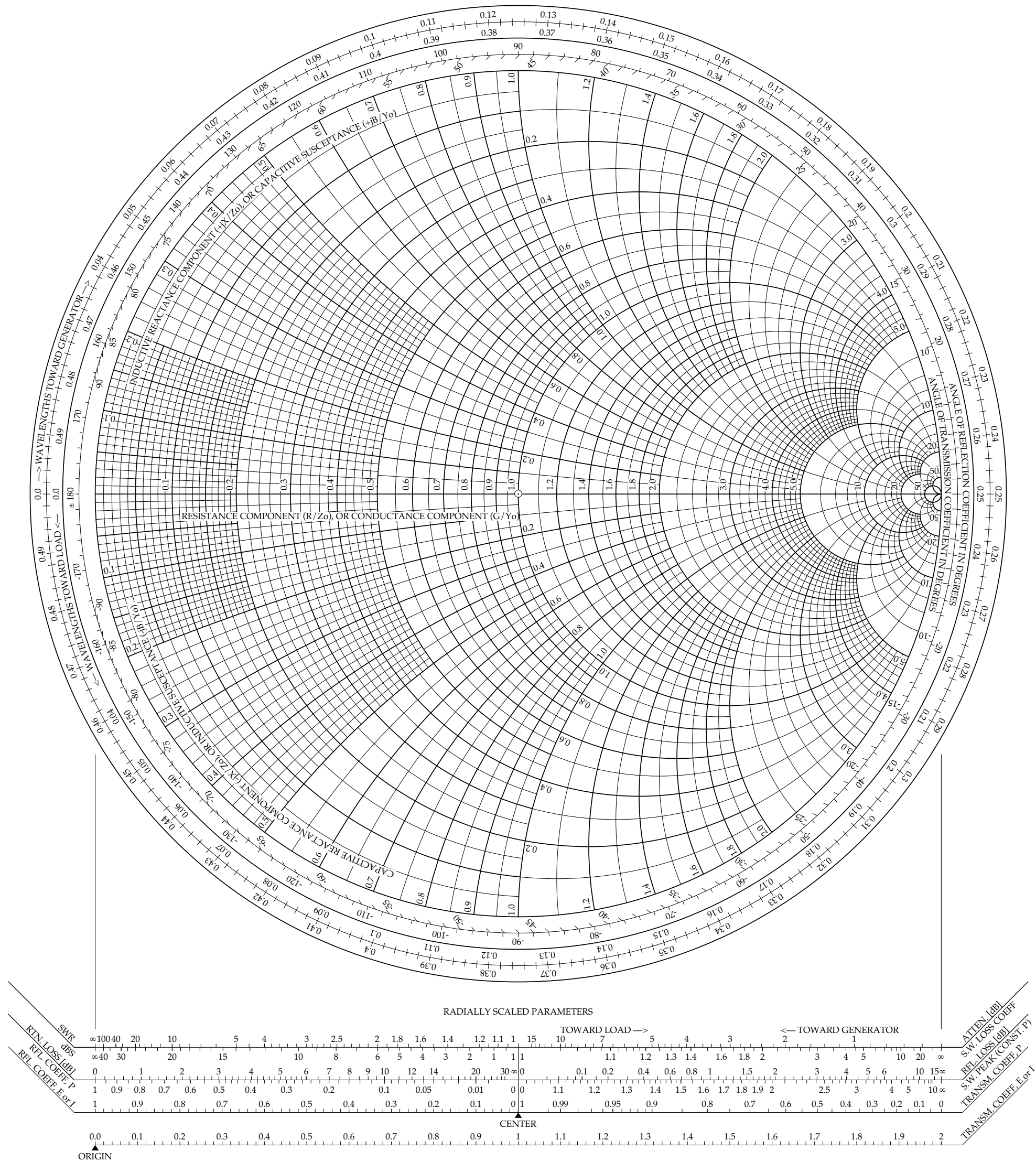
$$Y_L = -0.028j \quad X_C = -100j$$

Calculate actual values



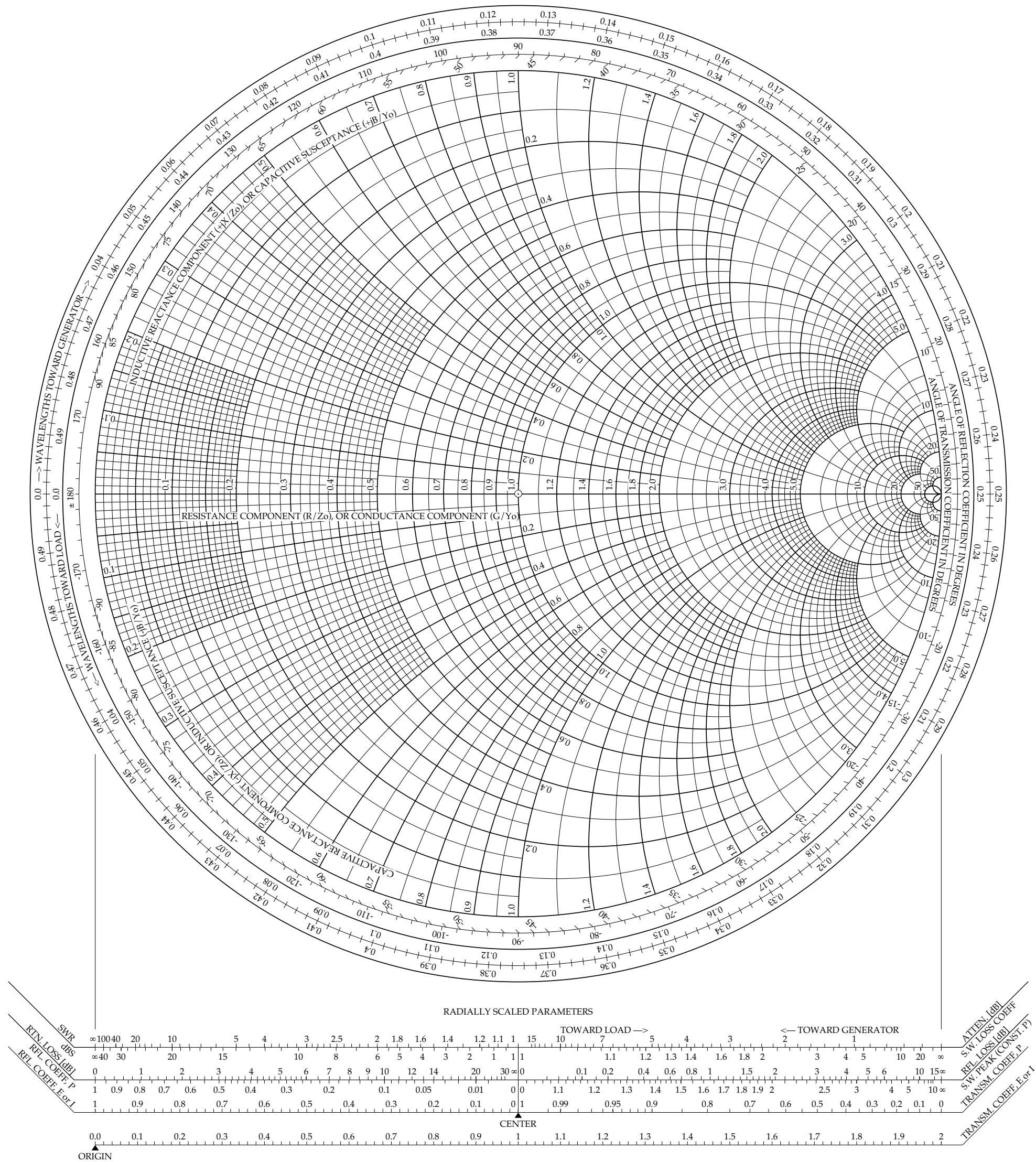
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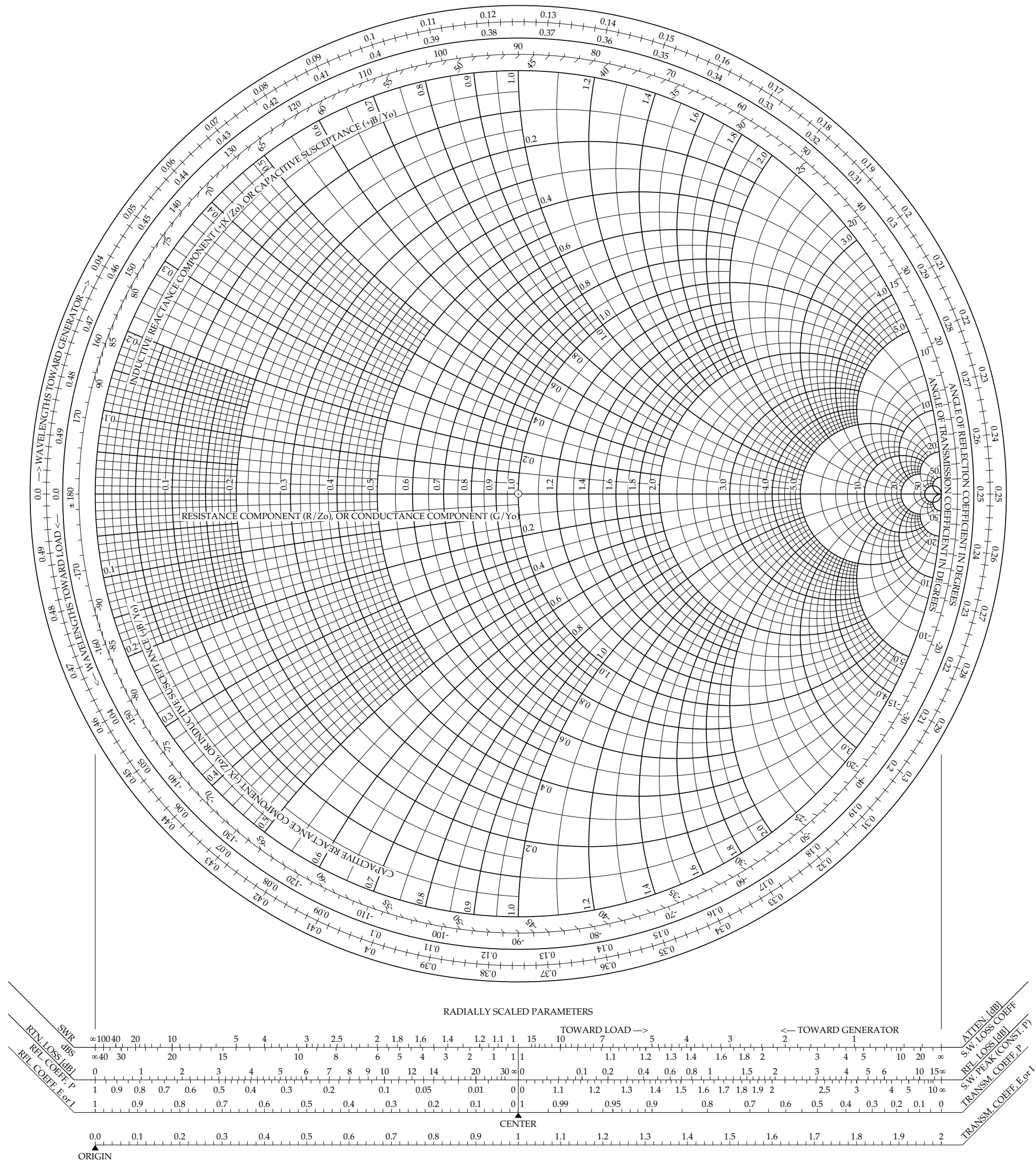
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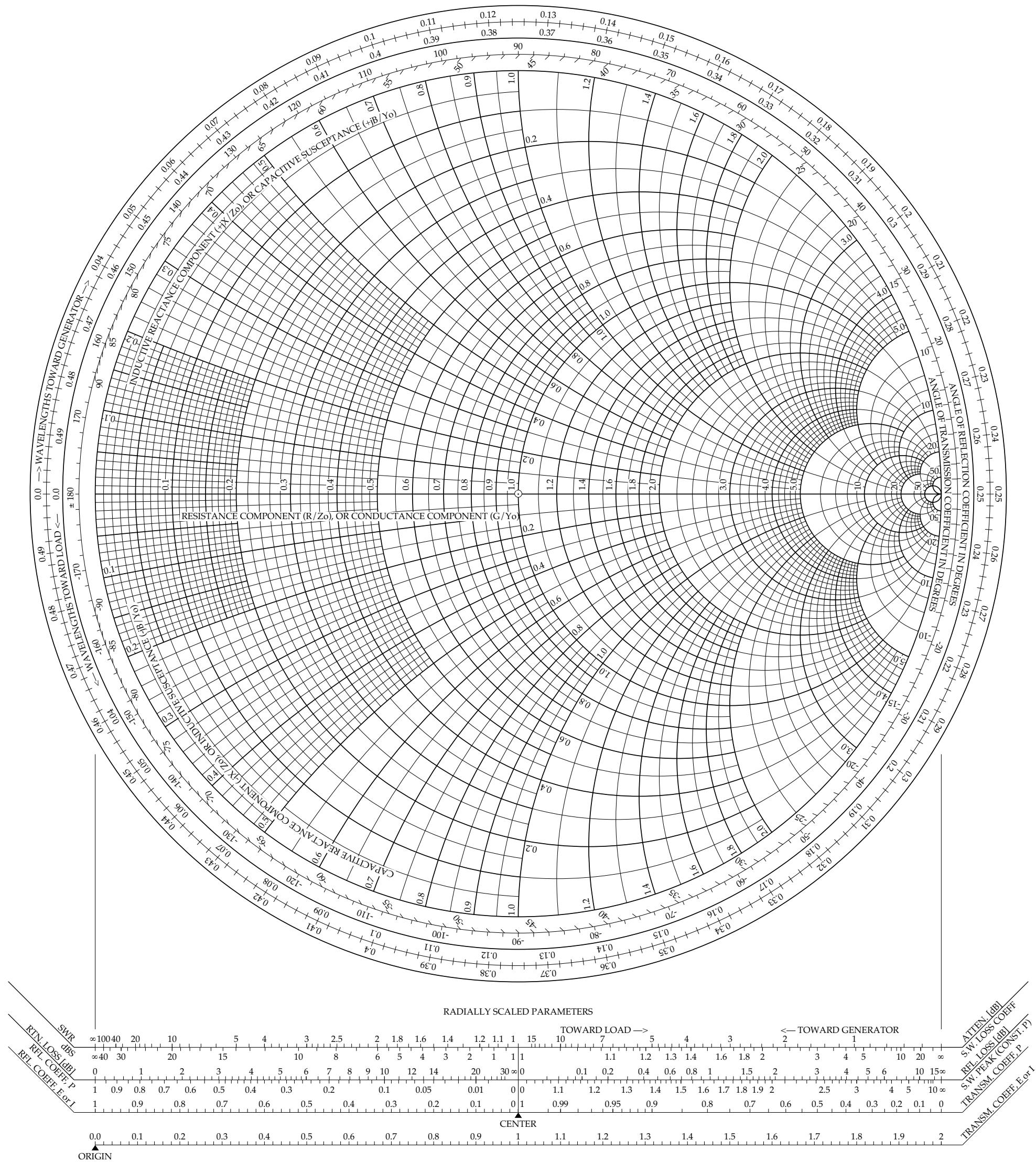
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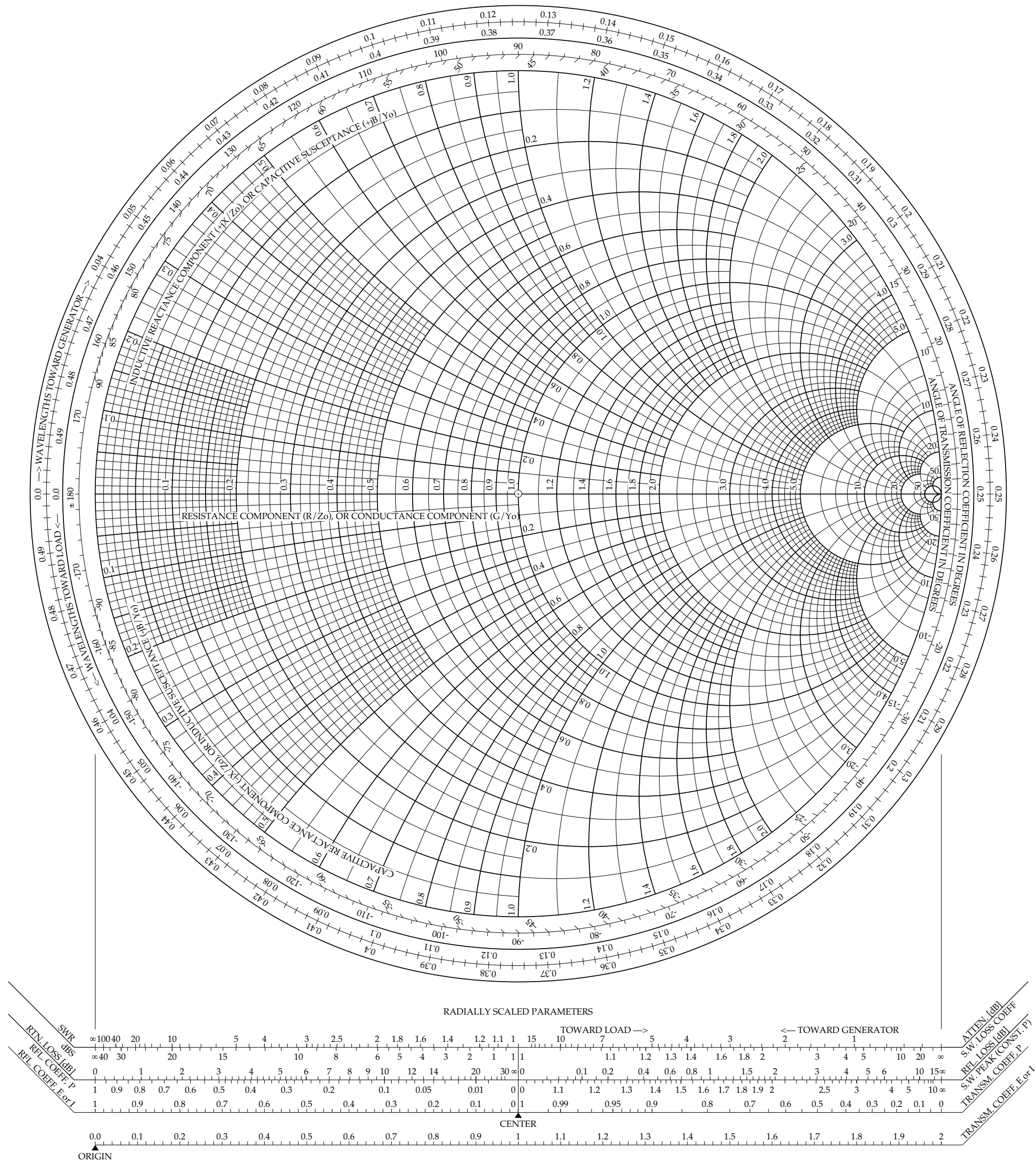
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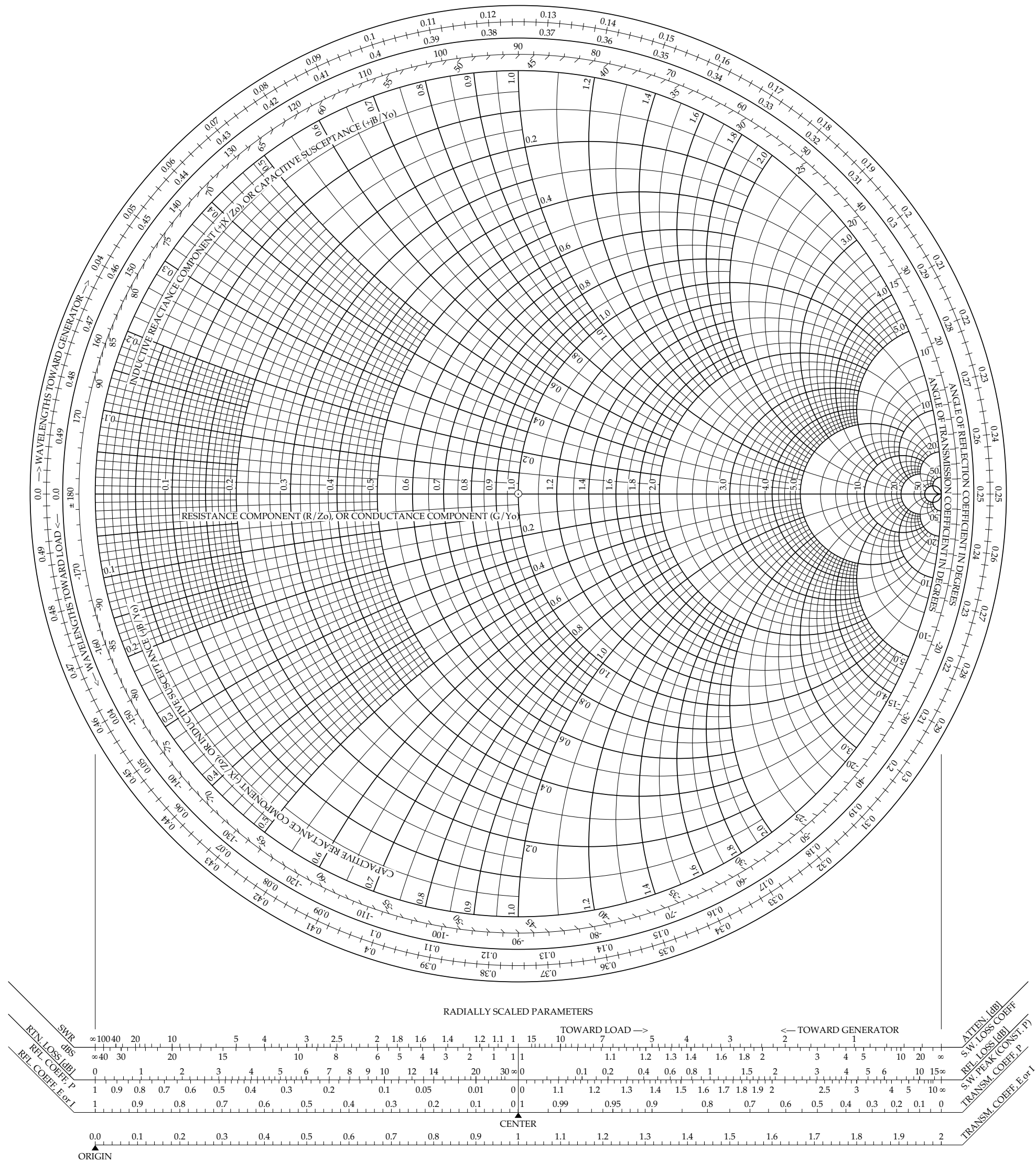
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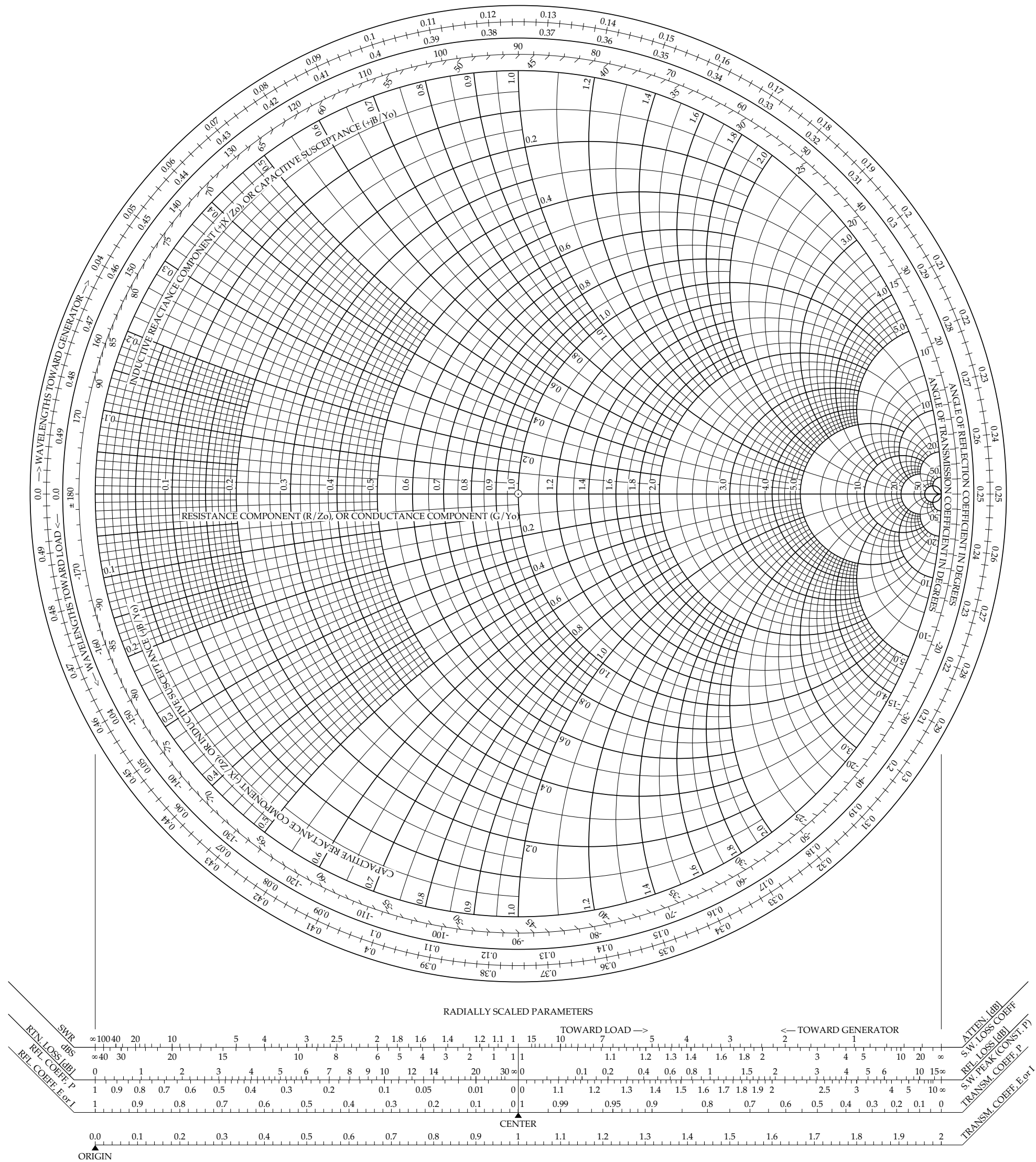
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